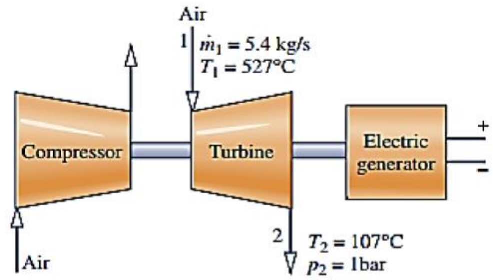
Figure shows a turbine operating at a steady-state that provides power to an air compressor and an electric generator. Air enters the turbine with a mass flow rate of 5.4 kg/s at 527 0C and exits the turbine at 1070C, 1 bar. The turbine provides power at a rate of 900 kW to the compressor and at a rate of 1400 kW to the generator. Air can be modeled as an ideal gas, and kinetic and potential energy changes are negligible. Determine (a) the volumetric flow rate of the air at the turbine exit, in m3 /s, and (b) the rate of heat transfer between the turbine and its surroundings, in kW.



Steady state

ΔKE and $\Delta PE = \emptyset$

$W_1 = 900KW \qquad W_2 = 1400KW$

a) $(AV)_2 = \dot{m}_2 v_2 = \dot{m}_2 \left[\frac{\bar{R} T_2}{p_2}\right]$

$$(AV)_2 = 5.4kg/s \left(\frac{\left(\frac{8314 N m/Kmol K}{28.97 Kg/Kmol}\right)(380K)}{1 bar (10^5 N/m^2)} \right)$$

$$= 5.88\ m^3/s$$

$$\boxed{Q_2 = 5.88\ m^3/s}$$

b) $\emptyset = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}\left[(h_1 - h_2) + \cancel{\frac{V_1^2 - V_2^2}{2}} + \cancel{g(z_1 - z_2)}\right]$

$\Rightarrow \dot{W}_{cv} = W_1 + W_2$

$= 900KW + 1400KW$

$= 2300KW$

$\Rightarrow$ find h_1 and h_2 from table A-22

@ $T_1 = 527°C = 800K \qquad h_1 = 821.95\ KJ/Kg$

@ $T_2 = 107°C = 380K \qquad h_2 = 380.77\ KJ/Kg$

$-\dot{Q}_{cv} = -\dot{W}_{cv} + \dot{m}(h_1 - h_2)$

$\dot{Q}_{cv} = \dot{W}_{cv} - \dot{m}(h_1 - h_2)$

$= (2300KW) - 5.4Kg/s\,(821.95\ KJ/Kg - 380.77\ KJ/Kg)$

$$\boxed{\dot{Q}_{cv} = -82.37\ KW}$$